torch.nn.functional.normalize#

https://pytorch.org/docs/stable/generated/torch.nn.functional.normalize.html

Description:

Perform LpL_pLp normalization of inputs over specified dimension. For a tensor input of sizes (n0,...,ndim,...,nk)(n_0, ..., n_{dim}, ..., n_k)(n0 ,...,ndim,...,nk), each ndimn_{dim}ndim -element vector vvv along dimension dim is transformed as With the default arguments it uses the Euclidean norm over vectors along dimension 111 for normalization. input (Tensor) – input tensor of any shape p (float) – the exponent value in the norm formulation. Default: 2

Function Signature

```
torch.nn.functional.normalize(input, p=2.0, dim=1, eps=1e-
12, out=None)[source]#

Parameters

Return type
```

Returns:

Tensor

Detailed Documentation

Documentation

Perform LpL_pLp normalization of inputs over specified dimension.

For a tensor input of sizes (n0,...,ndim,...,nk)(n_0, ..., n_{dim}, ..., n_k)(n0,...,ndim,...,nk), each ndimn_{dim}ndim -element vector vvv along dimension dim is transformed as

With the default arguments it uses the Euclidean norm over vectors along dimension 111 for normalization.

input (Tensor) – input tensor of any shape

p (float) – the exponent value in the norm formulation. Default: 2

dim (int or tuple of ints) – the dimension to reduce. Default: 1

eps (float) – small value to avoid division by zero. Default: 1e-12

out (Tensor, optional) – the output tensor. If out is used, this operation won't be differentiable.